

学霸助手

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课后答案 | 课件 | 期末试卷

最专业的学习资料分享APP

第一章习题提要

1. 1

$$y_i = \mu + e_i \quad i = 1, 2, \dots, n$$

$$e_i \sim N(0, \sigma^2)$$

1. 2

$$(1), \begin{pmatrix} y_1 \\ \vdots \\ y_{n_1} \\ z_1 \\ \vdots \\ z_{n_2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 0 & 1 \\ \vdots & \vdots \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} + \begin{pmatrix} e_1 \\ \vdots \\ e_{n_1} \\ \varepsilon_1 \\ \vdots \\ \varepsilon_{n_2} \end{pmatrix}, e_i \sim N(0, \sigma^2), i = 1, \dots, n_1, \varepsilon_i \sim N(0, \sigma^2), i = 1, \dots, n_2$$

$$(2) e_i \sim N(0, \sigma_1^2), i = 1, \dots, n_1, \varepsilon_i \sim N(0, \sigma_2^2), i = 1, \dots, n_2$$

1. 3

$$\begin{pmatrix} y_{11} \\ y_{12} \\ y_{13} \\ y_{21} \\ y_{22} \\ y_{23} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mu \\ \alpha_2 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} e_{11} \\ e_{12} \\ e_{13} \\ e_{21} \\ e_{22} \\ e_{23} \end{pmatrix}, e_{ij} \sim N(0, \sigma^2), i = 1, 2, j = 1, 2, 3$$

1. 4

$$(1) \text{ 令 } z_{i1} = x_{i1}, z_{i2} = x_{i1}^2, z_{i3} = \ln x_{i2}$$

$$(2) \text{ 取对数 } \ln y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i1}^2 + \ln e_i$$

$$\text{令 } w_i = \ln y_i, z_{i1} = x_{i1}, z_{i2} = x_{i1}^2, \varepsilon_i = \ln e_i$$

$$(3) \text{ 变形为 } \ln(y_i^{-2} - 1) = \beta_0 + \beta_1 x_{i1} + e_i$$

$$\text{令 } w_i = \ln(y_i^{-2} - 1)$$

$$(4) \text{ 令 } z_{i1} = x_{i1} + x_{i2}, z_{i2} = e^{x_{i1} + x_{i2}}, z_{i3} = \ln(x_{i1}^2)$$

1. 5

$$y = \begin{pmatrix} y_{11} \\ \vdots \\ y_{1J} \\ y_{21} \\ \vdots \\ y_{2J} \\ \vdots \\ y_{I1} \\ \vdots \\ y_{IJ} \end{pmatrix}, \quad e = \begin{pmatrix} e_{11} \\ \vdots \\ e_{1J} \\ e_{21} \\ \vdots \\ e_{2J} \\ \vdots \\ e_{I1} \\ \vdots \\ e_{IJ} \end{pmatrix}, \quad \beta = \begin{pmatrix} \mu \\ \alpha_1 \\ \vdots \\ \alpha_I \\ \beta_1 \\ \vdots \\ \beta_J \end{pmatrix}, \quad X = \begin{pmatrix} \mathbf{1}_J & \mathbf{1}_J & \mathbf{0}_J & & \mathbf{0}_J & \mathbf{1}_J \\ \mathbf{1}_J & \mathbf{0}_J & \ddots & & \mathbf{0}_J & \mathbf{1}_J \\ \mathbf{1}_J & \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{1}_J & \mathbf{0}_J & \mathbf{0}_J & & \mathbf{1}_J & \mathbf{1}_J \end{pmatrix}$$

第三章习题提要

3. 1

$$(1) Y = \begin{pmatrix} \mathbf{1}_5 & & & \\ & \mathbf{1}_5 & & \\ & & \mathbf{1}_5 & \\ & & & \mathbf{1}_5 \end{pmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \end{pmatrix} + e$$

$$(2) XX' = \begin{pmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}, (XX')^{-1} = \begin{pmatrix} \frac{1}{5} & 0 & 0 & 0 \\ 0 & \frac{1}{5} & 0 & 0 \\ 0 & 0 & \frac{1}{5} & 0 \\ 0 & 0 & 0 & \frac{1}{5} \end{pmatrix}, XY = \begin{pmatrix} \sum_{j=1}^5 y_{1j} \\ \sum_{j=1}^5 y_{2j} \\ \sum_{j=1}^5 y_{3j} \\ \sum_{j=1}^5 y_{4j} \end{pmatrix}$$

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \end{pmatrix} = \begin{pmatrix} \frac{1}{5} \sum_{j=1}^5 y_{1j} \\ \frac{1}{5} \sum_{j=1}^5 y_{2j} \\ \frac{1}{5} \sum_{j=1}^5 y_{3j} \\ \frac{1}{5} \sum_{j=1}^5 y_{4j} \end{pmatrix}$$

$$(3) \hat{\sigma}^2 = \frac{RSS}{n-p} = \frac{1}{16} \sum_{i=1}^4 \sum_{j=1}^5 (y_{ij} - \bar{y}_i)^2, \text{Var}(\hat{\mu}_i) = \frac{\hat{\sigma}^2}{5} = \frac{1}{5 \times 16} \sum_{i=1}^4 \sum_{j=1}^5 (y_{ij} - \bar{y}_i)^2$$

3. 2

$$(1) \begin{pmatrix} 20.2 \\ 8.0 \\ 9.7 \\ 1.9 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{pmatrix}$$

$$(2) \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}' \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix}$$

$$\hat{\mu} = (X'X)^{-1}X'Y = \begin{pmatrix} 9.95 \\ 5 \\ 4.15 \\ 1.1 \end{pmatrix}$$

$$(3) \text{Var}(\hat{\mu}) = \sigma^2(X'X)^{-1} = \frac{1}{4}\sigma^2 I_4$$

(4) 每个 4 次, 共 16 次

3. 3

$$\text{一元: } \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \beta_1 + \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}, \quad \hat{\beta}_1 = \left(\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}' \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right)^{-1} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}' \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

$$\text{二元: } \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}$$

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \left(\begin{pmatrix} x_{11} & x_{12} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix}' \begin{pmatrix} x_{11} & x_{12} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix} \right)^{-1} \begin{pmatrix} x_{11} & x_{12} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix}' \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\sum_{i=1}^n x_{i2}^2 \sum_{i=1}^n x_{i1} y_i - \sum_{i=1}^n x_{i1} x_{i2} \sum_{i=1}^n x_{i2} y_i}{\sum_{i=1}^n x_{i1}^2 \sum_{i=1}^n x_{i2}^2 - \left(\sum_{i=1}^n x_{i1} x_{i2} \right)^2} \\ \frac{\sum_{i=1}^n x_{i1}^2 \sum_{i=1}^n x_{i2} y_i - \sum_{i=1}^n x_{i1} x_{i2} \sum_{i=1}^n x_{i1} y_i}{\sum_{i=1}^n x_{i1}^2 \sum_{i=1}^n x_{i2}^2 - \left(\sum_{i=1}^n x_{i1} x_{i2} \right)^2} \end{pmatrix}$$

3. 4

$$\hat{y} = 22.6486 + 0.2643x$$

3. 5

$$\hat{y} = -0.1047 + 0.9881x$$

3. 6

$$\begin{pmatrix} \hat{\theta} \\ \hat{\phi} \end{pmatrix} = \begin{pmatrix} \frac{1}{6}y_1 + \frac{1}{3}y_2 + \frac{1}{6}y_3 \\ -\frac{1}{5}y_2 + \frac{2}{5}y_3 \end{pmatrix}, \quad \text{Cov} \begin{pmatrix} \hat{\theta} \\ \hat{\phi} \end{pmatrix} = \sigma^2 \begin{pmatrix} 0.167 & 0 \\ 0 & 0.2 \end{pmatrix}$$

3. 7

$$(1) X = \begin{pmatrix} \mathbf{1}_m & \mathbf{0}_m \\ \mathbf{1}_m & \mathbf{1}_m \\ \mathbf{1}_n & -2\mathbf{1}_n \end{pmatrix},$$

$$X'X = \begin{pmatrix} 2m+n & m-2n \\ m-2n & m+4n \end{pmatrix}, \quad (X'X)^{-1} = \begin{pmatrix} \frac{m+4n}{m(m+13n)} & \frac{-m+2n}{m(m+13n)} \\ \frac{-m+2n}{m(m+13n)} & \frac{2m+n}{m(m+13n)} \end{pmatrix}$$

$$(2) \begin{pmatrix} \hat{\theta} \\ \hat{\phi} \end{pmatrix} = \begin{pmatrix} \frac{(m+4n)\sum_{i=1}^{2m+n} y_i + (-m+2n)\left(\sum_{i=1}^m y_{m+i} - 2\sum_{i=1}^n y_{2m+i}\right)}{m(m+13n)} \\ \frac{(-m+2n)\sum_{i=1}^{2m+n} y_i + (2m+n)\left(\sum_{i=1}^m y_{m+i} - 2\sum_{i=1}^n y_{2m+i}\right)}{m(m+13n)} \end{pmatrix}$$

$$(3) m=2n \text{ 时, } Cov \begin{pmatrix} \hat{\theta} \\ \hat{\phi} \end{pmatrix} = \sigma^2 \begin{pmatrix} \frac{m+4n}{m(m+13n)} & 0 \\ 0 & \frac{2m+n}{m(m+13n)} \end{pmatrix}$$

3. 9

$$\text{设 } E(a'Y) = \theta \Leftrightarrow a'\mathbf{1}\theta = \theta \Leftrightarrow a'\mathbf{1} = 1$$

$$Var(a'Y) = a'\sigma^2 I a = \sigma^2 a'a$$

问题转化为约束极小值问题

$$L(a) = a'a - 2\lambda(a'\mathbf{1} - 1)$$

$$\text{可求得 } a = \frac{1}{n}\mathbf{1}, \quad \hat{\theta} = \frac{1}{n}\mathbf{1}'Y = \bar{y}$$

$$Var(\hat{\theta}) = \frac{1}{n}\mathbf{1}'\sigma^2 \frac{1}{n}\mathbf{1} = \frac{1}{n}\sigma^2$$

3. 10

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \left((X_1 X_2)' (X_1 X_2) \right)^{-1} (X_1 X_2)' Y = \begin{pmatrix} X_1' X_1 & X_1' X_2 \\ X_2' X_1 & X_2' X_2 \end{pmatrix}^{-1} \begin{pmatrix} X_1' Y \\ X_2' Y \end{pmatrix}$$

$$Cov \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \sigma^2 \begin{pmatrix} X_1' X_1 & X_1' X_2 \\ X_2' X_1 & X_2' X_2 \end{pmatrix}^{-1}$$

利用四块求逆公式即可

3. 11

(1) 由于 $X'y = (X'X)\hat{\beta}$, 记 $X = (\mathbf{1}': \tilde{X})$, 得 $\begin{pmatrix} \mathbf{1}'y \\ \tilde{X}'y \end{pmatrix} = \begin{pmatrix} \mathbf{1}'X\hat{\beta} \\ \tilde{X}'X\hat{\beta} \end{pmatrix}$, $\mathbf{1}'y = \mathbf{1}'X\hat{\beta}$,

$$\text{故 } \sum_{i=1}^n (y_i - \hat{y}_i) = \mathbf{1}'(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{1}'(\mathbf{y} - X\hat{\beta}) = 0$$

$$\begin{aligned} (2) \quad \sum_{i=1}^n \hat{y}_i (y_i - \hat{y}_i) &= (\mathbf{y} - \hat{\mathbf{y}})' \hat{\mathbf{y}} = (\mathbf{y} - X\hat{\beta})' (X\hat{\beta}) = \mathbf{y}'(I - X(X'X)^{-1}X')X(X'X)^{-1}X'y \\ &= \mathbf{y}'(X(X'X)^{-1}X' - X(X'X)^{-1}X')\mathbf{y} = 0 \end{aligned}$$

3. 12

$$L(\beta, \lambda) = (\mathbf{y} - \beta)'(\mathbf{y} - \beta) + 2\lambda'A\beta = \mathbf{y}'\mathbf{y} - 2\mathbf{y}'\beta + \beta'\beta + 2\lambda'A\beta$$

$$\frac{\partial}{\partial \beta} L(\beta, \lambda) = -2\mathbf{y} + 2\beta + 2A'\lambda = 0, \text{ 得 } \hat{\beta} = \mathbf{y} - A'\hat{\lambda}$$

$$\text{左乘 } A, \text{ 得 } 0 = A\mathbf{y} - AA'\hat{\lambda} \Rightarrow \hat{\lambda} = (AA')^{-1}A\mathbf{y}$$

$$\text{故 } \hat{\beta} = \mathbf{y} - A'(AA')^{-1}A\mathbf{y}$$

3. 13

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 \\ \vdots \\ n \end{pmatrix} \theta + \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}, \quad \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}, \sigma^2 \begin{pmatrix} 1^2 & & 0 \\ & \ddots & \\ 0 & & n^2 \end{pmatrix} \right)$$

广义最小二乘估计 $\hat{\theta} = (X'\Sigma^{-1}X)^{-1}(X'\Sigma^{-1}Y) = \sum_{i=1}^n \frac{y_i}{i}$ 是最小方差线性无偏估计。

$$\text{Var}(\hat{\theta}) = \sigma^2 (X'\Sigma^{-1}X)^{-1} = \sigma^2$$

3. 14

注意到 $\mathbf{1}'X_c = 0$

$$(a) \quad E(\hat{\alpha}) = E\left(\frac{1}{n}\mathbf{1}'y\right) = \frac{1}{n}E\left(\mathbf{1}'(\mathbf{1}\alpha + X_c\beta + e)\right) = \frac{1}{n}(\mathbf{1}'(\mathbf{1}\alpha + X_c\beta)) = \alpha + \frac{1}{n}\mathbf{1}'X_c\beta = \alpha$$

$$E(\hat{\beta}) = E\left(\left(X_c'X_c\right)^{-1}X_c'(\mathbf{1}\alpha + X_c\beta + e)\right) = \left(X_c'X_c\right)^{-1}X_c'(\mathbf{1}\alpha + X_c\beta) = \beta$$

(b)

$$\begin{aligned} \text{Cov}\begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \end{pmatrix} &= \text{Cov}\left(\begin{pmatrix} \frac{1}{n}\mathbf{1}' \\ \left(X_c'X_c\right)^{-1}X_c' \end{pmatrix} Y\right) = \begin{pmatrix} \frac{1}{n}\mathbf{1}' \\ \left(X_c'X_c\right)^{-1}X_c' \end{pmatrix} \sigma^2 I \begin{pmatrix} \frac{1}{n}\mathbf{1} & X_c\left(X_c'X_c\right)^{-1'} \end{pmatrix} \\ &= \sigma^2 \begin{pmatrix} \frac{1}{n} & 0 \\ 0 & \left(X_c'X_c\right)^{-1} \end{pmatrix} \end{aligned}$$

$$(c) \hat{\alpha} = \frac{1}{n}\mathbf{1}'y = \frac{1}{n}\left(\mathbf{1}'(\mathbf{1}\alpha + X_c\beta + e)\right) = \alpha + \frac{1}{n}\mathbf{1}'e \sim N\left(0, \sigma^2 \frac{1}{n}\mathbf{1}'\mathbf{1}\frac{1}{n}\right) = N\left(0, \frac{\sigma^2}{n}\right)$$

$$\hat{\beta} = \left(X_c'X_c\right)^{-1}X_c'\left(\mathbf{1}\alpha + X_c\beta + e\right) = \beta + \left(X_c'X_c\right)^{-1}X_c'e \sim N\left(\beta, \sigma^2\left(X_c'X_c\right)^{-1}\right)$$

3. 15

$$(1) \hat{y} = -0.8313 + 0.0037x$$

(2) 呈喇叭状

$$(1) \hat{u} = 0.5822 + 0.001x$$

(2) $\lambda = 0.5$ 时 $RSS = 96.97797$ 最小

3. 16

Model Summary(b)

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.973(a)	.947	.944	3.8990

a Predictors: (Constant), X2, X1

b Dependent Variable: Y

ANOVA(b)

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	7680.420	2	3840.210	252.607	.000(a)
	Residual	425.664	28	15.202		
	Total	8106.084	30			

a Predictors: (Constant), X2, X1

b Dependent Variable: Y

Coefficients(a)

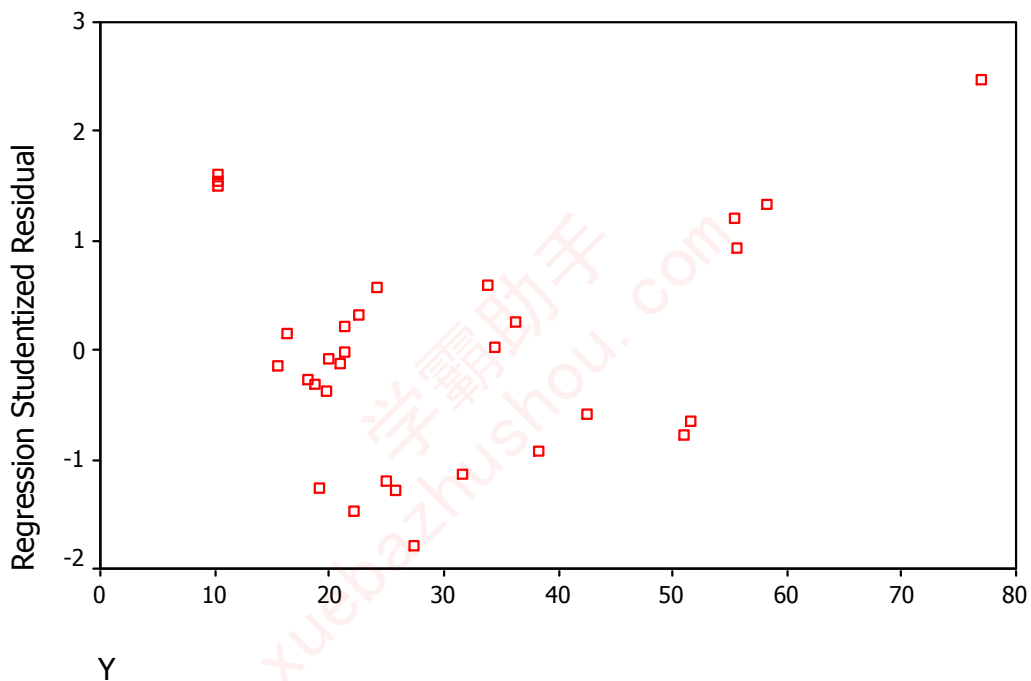
Model		Unstandardized	Standardized	t	Sig.
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		Coefficients		Coefficients		
		B	Std. Error	Beta		
1	(Constant)	-58.415	8.672		-6.736	.000
	X1	4.698	.265	.897	17.731	.000
	X2	.346	.131	.134	2.654	.013

a. Dependent Variable: Y

Scatterplot

Dependent Variable: Y



$$(1) \hat{y} = -58.415 + 4.698x_1 + 0.346x_2, \quad RSS = 425.664, \quad R = 0.973$$

残差图有 U 型趋势

计算比较可知 $\lambda = 0.25$ 时最优, $RSS = 137.044$

(2)

$$\frac{\hat{y}^{0.25} - 1}{0.25} = -1.987 + 0.341x_1 + 0.034x_2$$

3. 17

$$\hat{y} = -27.974 + 0.168x_1^2 + 0.355x_2, \quad RSS = 221.467, \quad R = 0.986, \quad \text{残差图均匀}$$

3. 18

$$(1) E(\hat{\beta}) = E((X'X)^{-1}X'Y) = (X'X)^{-1}X'X\beta = \beta$$

(2)

$$Cov(\hat{\beta}) = Cov((X'X)^{-1}X'Y) = (X'X)^{-1}X'Cov(Y)X(X'X)^{-1} = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1}$$

(3)

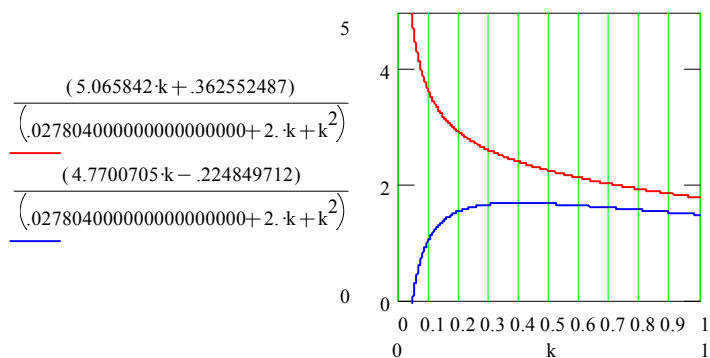
$$\begin{aligned} E(\hat{\sigma}^2) &= \frac{1}{n-p} E(y'(I - X(X'X)^{-1}X')y) \\ &= \frac{1}{n-p} (\beta'X(I - X(X'X)^{-1}X')X\beta) + \frac{\sigma^2}{n-p} tr((I - X(X'X)^{-1}X')V) \\ &= \frac{1}{n-p} (\beta'X(X\beta' - X\beta')) + \frac{\sigma^2}{n-p} tr(V(I - X(X'X)^{-1}X')) \\ &= \frac{\sigma^2}{n-p} tr(V(I - X(X'X)^{-1}X')) \end{aligned}$$

3. 19

$$(1) \text{ 中心化、标准化后 } X'X = \begin{pmatrix} 1 & 0.986 \\ 0.986 & 1 \end{pmatrix}$$

特征根 $\lambda_1 = 1.986$, $\lambda_2 = 0.014$, $\frac{\lambda_1}{\lambda_2} = 141.85$, 复共线性不严重。

$$\begin{aligned} (2) \hat{\beta}_1(k) &= \frac{5.065842k + 0.36255}{0.0278 + 2k + k^2} \\ \hat{\beta}_2(k) &= \frac{4.77k - 0.22485}{0.0278 + 2k + k^2} \end{aligned}$$



大致取 $k = 0.3$ 较好。

3. 20

直接作线性回归，所得经验回归方程为

$$\tilde{Y} = -17.564 + .091X_1 - 0.224X_2 + 0.015X_3 + 0.413X_4$$

下面作主成分回归：

原始数据： $X =$

82.90	92.00	17.10	94.00	8.40
88.00	93.00	21.30	96.00	9.60
99.00	96.00	25.10	97.00	10.40
105.30	94.00	29.00	97.00	11.40
117.70	100.00	34.00	100.00	12.20
131.00	101.00	40.00	101.00	14.20
148.20	105.00	44.00	104.00	15.80
161.80	112.00	49.00	109.00	17.90
174.20	112.00	51.00	111.00	19.60
184.70	112.00	53.00	111.00	20.80

中心化数据： $X_c =$

-46.38	-9.70	-19.25	-8.00
-41.28	-8.70	-15.05	-6.00
-30.28	-5.70	-11.25	-5.00
-23.98	-7.70	-7.35	-5.00
-11.58	-1.70	-2.35	-2.00
1.72	-.70	3.65	-1.00
18.92	3.30	7.65	2.00
32.52	10.30	12.65	7.00
44.92	10.30	14.65	9.00
55.42	10.30	16.65	9.00

$X_c'X_c =$

1.199E+4	2.616E+3	4.201E+3	2.08E+3
2.616E+3	594.1	917.75	465
4.201E+3	917.75	1.507E+3	723.9
2.08E+3	465	723.9	370

特征值： $\lambda_1 = 14397.7468$ ， $\lambda_2 = 34.26525$ ， $\lambda_3 = 25.57565$ ， $\lambda_4 = 2.21329$

$$\varphi_1 = \begin{pmatrix} -0.912 \\ -0.2 \\ -0.32 \\ -0.158 \end{pmatrix}, \quad \varphi_2 = \begin{pmatrix} -0.138 \\ -0.398 \\ 0.826 \\ -0.375 \end{pmatrix}, \quad \varphi_3 = \begin{pmatrix} 0.371 \\ -0.766 \\ -0.438 \\ -0.288 \end{pmatrix}, \quad \varphi_4 = \begin{pmatrix} -0.103 \\ -0.463 \\ 0.153 \\ 0.867 \end{pmatrix}$$

对应的前三个主成分

$$Z_1 = -0.912X_{c1} - 0.2X_{c2} - 0.32X_{c3} - 0.158X_{c4}$$

$$Z_2 = -0.138X_{c1} - 0.398X_{c2} + 0.826X_{c3} - 0.375X_{c4}$$

$$Z_3 = 0.371X_{c1} - 0.766X_{c2} - 0.438X_{c3} - 0.288X_{c4}$$

$$\text{回归得: } \hat{y} = 14.030 - 0.108Z_1 - 0.065Z_2 - 0.080Z_3$$

回代:

$$\hat{y} = 14.03 + 0.077786X_{c1} + 0.10875X_{c2} + 0.015910X_{c3} + 0.064479X_{c4}$$

$$\hat{y} = -14.2412 + 0.077786X_1 + 0.10875X_2 + 0.015910X_3 + 0.064479X_4$$

第二章习题提要

2. 1

$$Etr(AX) = E\left(\sum_{i=1}^p \sum_{k=1}^n a_{ik} x_{ki}\right)$$

$$trE(AX) = \sum_{i=1}^p E\left(\sum_{k=1}^n a_{ik} x_{ki}\right)$$

$$tr(AE(X)) = \sum_{i=1}^p \left(\sum_{k=1}^n a_{ik} Ex_{ki}\right)$$

2. 2

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ & \ddots & \ddots \\ 0 & 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix}$$

$$(1) \text{Cov}(Y) = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ & \ddots & \ddots \\ 0 & 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ & \ddots & \ddots \\ 0 & 0 & -1 & 1 \end{pmatrix}' = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & \ddots & 0 \\ & \ddots & \ddots & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

$$(2) \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ & \ddots & \ddots \\ 0 & 0 & -1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ & \vdots & \ddots \\ 1 & 1 & \cdots & 1 \end{pmatrix} \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}$$

$$\text{Cov}(X) = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ & \ddots & \ddots \\ 0 & 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ & \vdots & \ddots \\ 1 & 1 & \cdots & 1 \end{pmatrix}' = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2 & \ddots & 2 \\ 1 & 2 & 3 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & n-1 \\ 1 & 2 & \cdots & n-1 & n \end{pmatrix}$$

2. 3

$$(1) \text{Cov}(X+Y, X-Y) = E((X+Y)(X-Y)) - E(X+Y)E(X-Y) \\ = E(X^2) - E(Y^2) - E^2(X) + E^2(Y) = D(X) - D(Y) = 0$$

$$(2) \text{Cov}(X, Y - AX) = \text{Cov}(X, Y) - \text{Cov}(X, AX) = \text{Cov}(X, Y) - \text{Cov}(X)A'$$

$$\text{令 } \text{Cov}(X, Y) - \text{Cov}(X)A' = 0$$

$$\text{得 } A = \text{Cov}(X, Y)^T \text{Cov}(X)^{-1}$$

2. 4

$$(1) \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} = \begin{pmatrix} 1 & -1 & & 0 \\ 0 & 1 & -1 & 0 \\ & \ddots & \ddots & \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix}$$

$$E \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_{n-1} \end{pmatrix} = \begin{pmatrix} 1 & -1 & & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ & \ddots & \ddots & \ddots & \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix} \cdot E \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix} = \theta \begin{pmatrix} 1 & -1 & & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ & \ddots & \ddots & \ddots & \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$Cov \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_{n-1} \end{pmatrix} = \sigma^2 \begin{pmatrix} 1 & -1 & & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ & \ddots & \ddots & \ddots & \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 & & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ & \ddots & \ddots & \ddots & \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}^T = \sigma^2 \begin{pmatrix} 2 & -1 & & 0 \\ -1 & 2 & -1 & 0 \\ & -1 & \ddots & \ddots \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

(2)

$$E(Q) = \sum_{i=1}^{n-1} E((X_i - X_{i+1})^2) = \sum_{i=1}^{n-1} D((X_i - X_{i+1})) + \sum_{i=1}^{n-1} (E(X_i - X_{i+1}))^2 = 2(n-1)\sigma^2$$

2. 5

$$f(x_1, x_2, \dots, x_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_{ii}}} e^{-\frac{(x_i - \mu_i)^2}{2\sigma_{ii}}}$$

2. 8

$$\bar{X} = \left(\frac{1}{n}, \dots, \frac{1}{n} \right) X = BX$$

$$\sum_{i=1}^{n-1} (X_i - X_{i+1})^2 = X' C' C X$$

$$\text{其中 } C = \begin{pmatrix} 1 & -1 & \cdots & 0 \\ 0 & 1 & -1 & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & & & 1 & -1 \end{pmatrix}$$

$$BC' = 0, \text{ 故 } BC'C = 0$$

2. 9

$$Cov(U, V+W) = Cov(U, V) + Cov(U, W) = 0$$

2. 10

$$E(Y'AY) = \beta'X'AX\beta + \sigma^2 tr(AV)$$

须满足 $\begin{cases} X'AX = 0 \\ \text{tr}(AV) = 1 \end{cases}$

2. 11

$$\text{Cov}(Z) = \text{Cov}(AX + BY, AX + BY)$$

2. 12

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \sim N\left(\begin{pmatrix} 5 \\ 1 \end{pmatrix}, \begin{pmatrix} 10 & 0 \\ 0 & 3 \end{pmatrix}\right)$$

2. 13

$$\text{由} \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} 1 & \rho & 0 \\ \rho & 1 & \rho \\ 0 & \rho & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & -1 \end{pmatrix} = I$$

$$\text{得} \begin{pmatrix} 3+4\rho & -2\rho-1 \\ -2\rho-1 & 3 \end{pmatrix} = I, \quad \rho = -\frac{1}{2}$$

2. 14

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} X \sim N\left(\begin{pmatrix} \alpha'\mu \\ \beta'\mu \end{pmatrix}, \begin{pmatrix} \alpha'\alpha & \alpha'\beta \\ \beta'\alpha & \beta'\beta \end{pmatrix}\right)$$

2. 15

存在正交阵 Q 及 r 阶对角阵 Λ 使

$$A = Q \begin{pmatrix} \Lambda & 0 \\ 0 & 0 \end{pmatrix} Q' = (Q_1 : Q_2) \begin{pmatrix} \Lambda & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} Q_1' \\ Q_2' \end{pmatrix} = Q_1 \Lambda Q_1' = \left(Q_1 \Lambda^{\frac{1}{2}}\right) \left(Q_1 \Lambda^{\frac{1}{2}}\right)'$$

$$A \Sigma A = A \Leftrightarrow Q_1 \Lambda Q_1' \Sigma Q_1 \Lambda Q_1' = Q_1 \Lambda Q_1' \Leftrightarrow \Lambda^{\frac{1}{2}} Q_1' \Sigma Q_1 \Lambda^{\frac{1}{2}} = I_r$$

$$(X - \mu)' A (X - \mu) = \left(\Lambda^{\frac{1}{2}} Q_1' (X - \mu) \right)' \left(\Lambda^{\frac{1}{2}} Q_1' (X - \mu) \right)$$

$$\text{而} \left(\Lambda^{\frac{1}{2}} Q_1' (X - \mu) \right) \sim N_r \left(0, \Lambda^{\frac{1}{2}} Q_1' \Sigma Q_1 \Lambda^{\frac{1}{2}} \right) = N_r(0, I_r)$$

2. 16

由于 Σ 对称非负定, 存在 $Q_{n \times r}$ 使 $\Sigma = Q Q'$, 其中 $r = R(\Sigma)$

记 $U \sim N_r(0, I)$, 则 $X = \mu + Q U$

$$(X - \mu)' A(X - \mu) = U' (Q' A Q) U$$

$$(X - \mu)' B(X - \mu) = U' (Q' B Q) U$$

$$(Q' A Q)(Q' B Q) = 0 \Leftrightarrow Q' A \Sigma B Q = 0 \Leftrightarrow Q Q' A \Sigma B Q Q' = 0 \Leftrightarrow \Sigma A \Sigma B \Sigma = 0$$

2. 17

$$(x_1^2 + 2x_2^2 - x_1 x_2 - 3x_1 - 2x_2 + 4) = (x_1 - 2, x_2 - 1) \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 2 \end{pmatrix} \begin{pmatrix} x_1 - 2 \\ x_2 - 1 \end{pmatrix}$$

$$\text{故 } E(X) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \text{Cov}(X) = \begin{pmatrix} \frac{8}{7} & \frac{2}{7} \\ \frac{2}{7} & \frac{4}{7} \end{pmatrix}$$

2. 18

$$E(e^{it'X}) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} e^{it'x} \frac{1}{(\sqrt{2\pi})^p |\Sigma|} \exp\left(-\frac{1}{2}(x - \mu)' \Sigma^{-1}(x - \mu)\right) dx_1 \cdots dx_p$$

$$\text{作变换 } y = \Sigma^{\frac{1}{2}}(x - \mu),$$

$$\text{则 } x = \Sigma^{\frac{1}{2}}y + \mu$$

$$\text{变换的 Jacobian 行列式 } J = \left| \Sigma^{\frac{1}{2}} \right| = |\Sigma|^{\frac{1}{2}}$$

$$\text{用 } c_k \text{ 表示 } \Sigma^{\frac{1}{2}} \text{ 的第 } k \text{ 列, 则 } \Sigma^{\frac{1}{2}} = (c_1, c_2, \cdots, c_p),$$

$$\text{由于 } \Sigma^{\frac{1}{2}} \text{ 是对称矩阵, 所以又有行表示 } \Sigma^{\frac{1}{2}} = \begin{pmatrix} c'_1 \\ c'_2 \\ \vdots \\ c'_p \end{pmatrix},$$

$$\Sigma = \Sigma^{\frac{1}{2}} \Sigma^{\frac{1}{2}} = \begin{pmatrix} c_1 & c_2 & \cdots & c_p \end{pmatrix} \begin{pmatrix} c'_1 \\ c'_2 \\ \vdots \\ c'_p \end{pmatrix} = \sum_{k=1}^p c_k c'_k$$

$$E(e^{it'X}) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} e^{it'(\Sigma^{\frac{1}{2}}y + \mu)} \frac{1}{(\sqrt{2\pi})^p} \exp\left(-\frac{1}{2}y' y\right) dy_1 \cdots dy_p$$

$$= e^{it'\mu} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} e^{it'(\Sigma^{\frac{1}{2}}y)} \frac{1}{(\sqrt{2\pi})^p} \exp\left(-\frac{1}{2}y' y\right) dy_1 \cdots dy_p$$

$$= e^{it'\mu} \prod_{k=1}^p \int_{-\infty}^{\infty} \left(e^{i(t'c_k)y_k} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y_k^2\right) \right) dy_k$$

$$= e^{it'\mu} \prod_{k=1}^p e^{-\frac{1}{2}(t'c_k)^2}$$

$$= e^{it'\mu} \prod_{k=1}^p e^{-\frac{1}{2}(t'c_k c'_k t)}$$

$$= e^{it'\mu} e^{-\frac{1}{2}t' \left(\sum_{k=1}^p c_k c'_k \right) t}$$

$$= e^{it'\mu} e^{-\frac{1}{2}t'\Sigma t}$$

2. 19

($\Rightarrow$) 利用习题 2.18 的结论

$a'X$ 的特征函数

$$E(e^{ita'X}) = E(e^{i(ta)'X}) = e^{i(ta)'\mu - \frac{1}{2}(ta)'\Sigma(ta)} = e^{it(a'\mu) - \frac{1}{2}t(a'\Sigma a)t}$$

这正是一元正态分布 $N(a'\mu, a'\Sigma a)$ 的特征函数。

($\Leftarrow$) 设 $E(X) = \mu$, $Cov(X) = \Sigma$, 若对任意的实向量 $a (a \neq 0)$, $a'X$ 是一元正态分布,

则 $a'X \sim N(a'\mu, a'\Sigma a)$, $a'X$ 的特征函数

$$E(e^{ita'X}) = e^{it(a'\mu) - \frac{1}{2}t(a'\Sigma a)t}$$

取 $t = 1$, 得 $E(e^{ia'X}) = e^{i(a'\mu) - \frac{1}{2}a'\Sigma a}$ 这正是随机向量 X 的特征函数在 a 处的值。由于 a 的任意性, 表明 X 的特征函数与 $N_p(\mu, \Sigma)$ 的特征函数完全相同, 由特征函数与分

布的相互唯一确定性得证 $X \sim N_p(\mu, \Sigma)$

2. 20

$$(x^2 + xy + 3y^2) = (x, y) \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$-(x^2 + 2xy + 2y^2) = -\frac{1}{2}(x-0, y-0) \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}^{-1} \begin{pmatrix} x-0 \\ y-0 \end{pmatrix}$$

$$2\pi\sqrt{|\Sigma|} = 2\pi\sqrt{\frac{1}{4}} = \pi$$

$$\text{即} \begin{pmatrix} X \\ Y \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} \right)$$

$$\text{原式} = \pi E \left[(x, y) \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \right] = \pi \cdot \text{tr}(A\Sigma)$$

$$= \pi \cdot \text{tr} \left(\begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 3 \end{pmatrix} \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} \right) = \pi \cdot \text{tr} \left(\begin{pmatrix} \frac{3}{4} & -\frac{1}{4} \\ -1 & \frac{5}{4} \end{pmatrix} \right) = 2\pi$$

第五章习题提要

5. 1

第一步引入 x_{i_1} 意味着 x_{i_1} 的偏回归平方和 $p_{i_1}^{(1)} = \max(p_1^{(1)}, p_2^{(1)}, p_3^{(1)}, p_4^{(1)})$, 且

$$F_{\text{剔}}^{(1)}(i_1) = \frac{(n-q-1)p_{i_1}^{(1)}}{RSS_1(i_1)} > F_{1,n-q-1}(\alpha)$$

第二步引入 x_{i_2} 意味着 x_{i_2} 的偏回归平方和 $p_{i_2}^{(2)} = \max(p_{i_2}^{(2)}, p_{i_3}^{(2)}, p_{i_4}^{(2)})$, 且

$$F_{\text{剔}}^{(2)}(i_2) = \frac{(n-q-2)p_{i_2}^{(2)}}{RSS_2(i_1, i_2)} > F_{1,n-q-2}(\alpha)$$

第三步: 由于 $F_{\text{剔}}^{(3)}(i_2) = \frac{(n-q-2)(RSS_1(i_1) - RSS_2(i_1, i_2))}{RSS_2(i_1, i_2)}$

$$= \frac{(n-q-2)p_{i_2}^{(2)}}{RSS_2} > F_{1,n-q-2}(\alpha), \text{ 所以不能剔除 } x_{i_2}$$

又由于 $RSS_1(i_2) = a_{55}^{(0)} - p_{i_2}^{(1)} > a_{55}^{(0)} - p_{i_1}^{(1)} = RSS_1(i_1)$

$$F_{\text{剔}}^{(3)}(i_1) = \frac{(n-q-2)(RSS_1(i_2) - RSS_2(i_1, i_2))}{RSS_2(i_1, i_2)}$$

$$> \frac{(n-q-2)(RSS_1(i_1) - RSS_2(i_1, i_2))}{RSS_2(i_1, i_2)}$$

$$= \frac{(n-q-2)p_{i_2}^{(2)}}{RSS_2} > F_{1,n-q-2}(\alpha), \text{ 所以也不能剔除 } x_{i_1},$$

故第三步不能剔除变量。

5. 2

$$\text{记 } X_q = \begin{pmatrix} 1 & x_{1_1} \\ \vdots & \vdots \\ 1 & x_{n_1} \end{pmatrix}, \quad X_2 = \begin{pmatrix} x_{1_2} \\ \vdots \\ x_{n_2} \end{pmatrix}, \quad \beta_q = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$$

$$\text{则 } \hat{\sigma}^2 = \frac{RSS_q}{n-2} = y'(I - X_q(X_q'X_q)^{-1}X_q')y$$

$$\text{在全模型正确时, } y = \begin{pmatrix} 1 & x_{1_1} & x_{1_2} \\ \vdots & \vdots & \vdots \\ 1 & x_{n_1} & x_{n_2} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} + e = (X_q : X_2) \begin{pmatrix} \beta_q \\ \beta_2 \end{pmatrix} + e$$

$$= X_q\beta_q + X_2\beta_2 + e,$$

$$E(y) = X_q \beta_q + X_2 \beta_2, \quad \text{Cov}(y) = \sigma^2 I$$

$$\begin{aligned} E(\hat{\sigma}^2) &= E\left[y'(I_n - X_q(X_q'X_q)^{-1}X_q')y\right] \\ &= (X_q\beta_q + X_2\beta_2)'(I_n - X_q(X_q'X_q)^{-1}X_q')(X_q\beta_q + X_2\beta_2) \\ &\quad + \text{tr}\left[\sigma^2(I_n - X_q(X_q'X_q)^{-1}X_q')\right] \\ &= \beta_2'X_2'(I_n - X_q(X_q'X_q)^{-1}X_q')X_2\beta_2 + \sigma^2(n-2) \end{aligned}$$

计算结果表明 $\frac{RSS_q}{n-2}$ 不是 σ^2 的无偏估计，且偏大。

5. 3

变量子集	RSS_q	$RMS_q = \frac{1}{n-q} RSS_q$	$c_p = \frac{RSS_q}{\hat{\sigma}^2} - (n-2q)$ ($\hat{\sigma}^2 = \frac{5.516}{18-4} = 0.394$)	$AIC = n \ln(RSS_q) + 2q$
X_1	8.325	0.520	7.129	42.146
X_2	13.230	0.827	19.579	50.485
X_3	25.375	1.586	50.404	62.208
X_1, X_2	7.715	0.514	7.581	42.777
X_1, X_3	7.791	0.519	7.774	42.953
X_2, X_3	9.621	0.641	12.419	46.751
X_1, X_2, X_3	5.516	0.368*	4*	38.738*

可见，三种标准都以 X_1, X_2, X_3 为最优子集，

最优回归方程为 $y = -6.984 + 0.074X_1 + 0.462X_2 - 0.351X_3$

5. 4

取 $\alpha = 0.1$

逐步回归的结果为 $y = -0.673 + 0.108X_1$

向前法的结果也是 $y = -0.673 + 0.108X_1$

向后法的结果是 $y = -6.984 + 0.074X_1 + 0.462X_2 - 0.351X_3$

5. 5

取 $\alpha = 0.1$

逐步回归的结果为 $y = 1637.196 + 290.903X_3$

向前法的结果为 $y = 1637.196 + 290.903X_3$

向后法的结果是 $y = -4187.416 + 80.271X_1 + 46.449X_2$

意味着 X_1, X_2 组合起来，大致可取代 X_3 的作用。

5.6

变量子集	RSS_q	$RMS_q = \frac{1}{n-q} RSS_q$	$c_p = \frac{RSS_q}{\hat{\sigma}^2} - (n-2q)$ $(\hat{\sigma}^2 = \frac{445140.93}{10-4} = 74190.16)$	$AIC = n \ln(RSS_q) + 2q$
X_1	877477.16	109684.65	5.827	140.85
X_2	1112337.91	139042.24	8.993	143.22
X_3	794759.42	99344.93	4.712	139.86
X_1, X_2	496392.44	70913.21*	2.691*	137.15*
X_1, X_3	645953.43	92279.06	4.707	139.78
X_2, X_3	654210.03	93458.58	4.818	139.91
X_1, X_2, X_3	445140.93	74190.16	4	138.06

可见，三种标准都以 X_1, X_2 为最优子集，

最优回归方程为 $y = -4187.416 + 80.271X_1 + 46.449X_2$

5.7 (方法与 5.5 和 5.6 相同，但有 31 个子集要比较。从略)

第六章习题提要

6. 1

$$\sum_{i=1}^a n_i \alpha_i = \sum_{i=1}^a n_i (\mu_i - \mu) = \sum_{i=1}^a n_i \mu_i - \sum_{i=1}^a n_i \mu = n\mu - \sum_{i=1}^a n_i \mu = 0$$

6. 2

$$\mu_1 = \cdots = \mu_a \Leftrightarrow \mu_1 - \mu = \cdots = \mu_a - \mu \Leftrightarrow \alpha_1 = \cdots = \alpha_a$$

在条件 $\sum_{i=1}^a \alpha_i = 0$ 下

$$\Leftrightarrow \alpha_1 = \cdots = \alpha_a = 0$$

6. 3

$y_{ij} \sim N(\mu_i, \sigma^2), i=1,2,\cdots,a, j=1,2,\cdots,n_i$, 且相互独立。

$$\text{记 } X = \begin{pmatrix} \mathbf{1}_{n_1} & & & \\ & \mathbf{1}_{n_2} & & \\ & & \ddots & \\ & & & \mathbf{1}_{n_a} \end{pmatrix}, \beta = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_a \end{pmatrix}, Y = \begin{pmatrix} y_{11} \\ \vdots \\ y_{1n_1} \\ \vdots \\ y_{a1} \\ \vdots \\ y_{an_a} \end{pmatrix}, n = \sum_{i=1}^a n_i$$

则 $Y \sim N_a(X\beta, \sigma^2 I_n)$

$$\text{记 } e_i = \begin{pmatrix} \mathbf{0}_{n_1} \\ \vdots \\ \mathbf{1}_{n_i} \\ \vdots \\ \mathbf{0}_{n_1} \end{pmatrix}$$

$$\text{则 } \bar{y}_{i\cdot} - \bar{y}_{j\cdot} = \left(\frac{1}{n_i} e_i' - \frac{1}{n_j} e_j' \right) Y$$

$$\text{由于 } \left(\frac{1}{n_i} e_i' - \frac{1}{n_j} e_j' \right) X\beta = \mu_i - \mu_j$$

$$\begin{aligned} & \left(\frac{1}{n_i} e_i' - \frac{1}{n_j} e_j' \right) \sigma^2 \left(\frac{1}{n_i} e_i' - \frac{1}{n_j} e_j' \right)' = \left(\frac{1}{n_i} e_i' - \frac{1}{n_j} e_j' \right) \sigma^2 \left(\frac{1}{n_i} e_i - \frac{1}{n_j} e_j \right) \\ & = \sigma^2 \left(\frac{e_i' e_i}{n_i n_i} - 2 \frac{e_i' e_j}{n_i n_j} + \frac{e_j' e_j}{n_j n_j} \right) = \sigma^2 \left(\frac{1}{n_i} + \frac{1}{n_j} \right) \end{aligned}$$

$$\text{故 } \bar{y}_i - \bar{y}_j \sim N\left(\mu_i - \mu_j, \left(\frac{1}{n_i} + \frac{1}{n_j}\right)\sigma^2\right)$$

6. 4

$$(1) 40 = \mu_1 + e_{11}, 48 = \mu_1 + e_{12}, 38 = \mu_1 + e_{13}, \dots$$

$$26 = \mu_2 + e_{21}, \dots$$

$$39 = \mu_3 + e_{31}, \dots$$

(2) 有显著差异

ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	855.111	2	427.556	23.251	0.00002535
Within Groups	275.833	15	18.389		
Total	1130.944	17			

表明有显著差异

(3)

$\mu_A - \mu_B$ 的 95% 置信区间为 (7.39, 17.94)

$\mu_A - \mu_C$ 的 95% 置信区间为 (-8.61, 1.94)

$\mu_B - \mu_C$ 的 95% 置信区间为 (-21.28, -10.72)

Multiple Comparisons

Dependent Variable: Y

LSD

(I) A	(J) A	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
1	2	12.67(*)	2.476	.000	7.39	17.94
	3	-3.33	2.476	.198	-8.61	1.94
2	1	-12.67(*)	2.476	.000	-17.94	-7.39
	3	-16.00(*)	2.476	.000	-21.28	-10.72
3	1	3.33	2.476	.198	-1.94	8.61
	2	16.00(*)	2.476	.000	10.72	21.28

* The mean difference is significant at the .05 level.

表明:

$\mu_A - \mu_B$ 的 95% 置信区间为 (7.39, 17.94)

$\mu_A - \mu_C$ 的 95% 置信区间为 (-8.61, 1.94)

$\mu_B - \mu_C$ 的95%置信区间为 (-21.28, -10.72)

6.5

$$\sum_{i=1}^a \alpha_i = \sum_{i=1}^a (\bar{\mu}_{i.} - \mu) = \sum_{i=1}^a \bar{\mu}_{i.} - a\mu = \frac{1}{b} \sum_{i=1}^a \sum_{j=1}^b \mu_{ij} - a \frac{1}{ab} \sum_{i=1}^a \sum_{j=1}^b \mu_{ij} = 0$$

$$\sum_{j=1}^b \beta_j = \sum_{j=1}^b (\bar{\mu}_{.j} - \mu) = \sum_{j=1}^b \bar{\mu}_{.j} - b\mu = \frac{1}{a} \sum_{j=1}^b \sum_{i=1}^a \mu_{ij} - b \frac{1}{ab} \sum_{i=1}^a \sum_{j=1}^b \mu_{ij} = 0$$

6.6

(1)

$$50 = \mu + \alpha_1 + \beta_1 + e_{11}$$

$$47 = \mu + \alpha_1 + \beta_2 + e_{21}$$

....

$$48 = \mu + \alpha_3 + \beta_4 + e_{34}$$

(2)

ANOVA Table

		Sum of Squares	df	Mean Square	F	Sig.
Y * W	Between Groups	114.667	3	38.222	.870	.495
	Within Groups	351.333	8	43.917		
	Total	466.000	11			

表明四名工人的日加工量没有显著差异

(3)

ANOVA Table

		Sum of Squares	df	Mean Square	F	Sig.
Y * A	Between Groups	318.500	2	159.250	9.717	.006
	Within Groups	147.500	9	16.389		
	Total	466.000	11			

表明三台机床的日加工量存在显著差异

6.7

$y_{ij} \sim N(\mu + \alpha_i + \beta_j, \sigma^2), i=1,2,\dots,a, j=1,2,\dots,b$, 且相互独立。

记

$$X = \begin{pmatrix} \mathbf{1}_b & \mathbf{1}_b & & & \mathbf{I}_b \\ \vdots & & \ddots & & \vdots \\ \mathbf{1}_b & & & \mathbf{1}_b & \mathbf{I}_b \end{pmatrix}, \theta = \begin{pmatrix} \mu \\ \alpha_1 \\ \vdots \\ \alpha_a \\ \beta \end{pmatrix}, Y = \begin{pmatrix} y_1 \\ \vdots \\ y_a \end{pmatrix},$$

$$\text{其中 } \beta = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_b \end{pmatrix}, y_i = \begin{pmatrix} y_{i1} \\ \vdots \\ y_{ib} \end{pmatrix}$$

$$\text{则 } Y \sim N_a(X\theta, \sigma^2 I_{ab})$$

$$\text{记 } e_i = \begin{pmatrix} \mathbf{0}_b \\ \vdots \\ \mathbf{1}_b \\ \vdots \\ \mathbf{0}_b \end{pmatrix} \text{ 表示仅第 } i \text{ 子块全为 } \mathbf{1}$$

$$\text{则 } \bar{y}_{i\cdot} - \bar{y}_{t\cdot} = \left(\frac{1}{b} e'_i - \frac{1}{b} e'_t \right) Y$$

$$\text{由于 } \left(\frac{1}{b} e'_i - \frac{1}{b} e'_t \right) X\theta = \alpha_i - \alpha_t$$

$$\begin{aligned} & \left(\frac{1}{b} e'_i - \frac{1}{b} e'_t \right) \sigma^2 \left(\frac{1}{b} e'_i - \frac{1}{b} e'_t \right)' = \left(\frac{1}{b} e'_i - \frac{1}{b} e'_t \right) \sigma^2 \left(\frac{1}{b} e_i - \frac{1}{b} e_t \right) \\ & = \sigma^2 \frac{1}{b^2} (e'_i e_i - 2e'_i e_t + e'_t e_t) = \sigma^2 \frac{2}{b} \end{aligned}$$

$$\text{故 } \bar{y}_{i\cdot} - \bar{y}_{t\cdot} \sim N\left(\alpha_i - \alpha_t, \sigma^2 \frac{2}{b}\right)$$

6.8

$$\bar{y}_{i\cdot} - \bar{y}_{t\cdot} \sim N\left(\alpha_i - \alpha_t, \sigma^2 \frac{2}{b}\right)$$

$\frac{SS_e}{\sigma^2} \sim \chi^2((a-1)(b-1))$, 且与 $\bar{y}_{i\cdot} - \bar{y}_{t\cdot}$ 相互独立, 有 t 分布的定义

$$\begin{aligned} & \frac{(\bar{y}_{i\cdot} - \bar{y}_{t\cdot}) - (\alpha_i - \alpha_t)}{\sqrt{\frac{2}{b} \sqrt{\frac{SS_e}{(a-1)(b-1)}}}} \sim t((a-1)(b-1)) \\ & \Leftrightarrow \frac{(\bar{y}_{i\cdot} - \bar{y}_{t\cdot}) - (\alpha_i - \alpha_t)}{\sqrt{\frac{2}{b} \hat{\sigma}}} \sim t((a-1)(b-1)) \end{aligned}$$

$$P\left\{\left|\frac{(\bar{y}_i - \bar{y}_{t.}) - (\alpha_i - \alpha_{t.})}{\sqrt{\frac{2}{b}\hat{\sigma}^2}}\right| \leq t_{\frac{\alpha}{2}}((a-1)(b-1))\right\} = 1 - \alpha$$

$$\Leftrightarrow P\left\{\begin{aligned} &(\bar{y}_i - \bar{y}_{t.}) - \sqrt{\frac{2}{b}\hat{\sigma}^2} t_{\frac{\alpha}{2}}((a-1)(b-1)) \leq \alpha_i - \alpha_{t.} \\ &\leq (\bar{y}_i - \bar{y}_{t.}) + \sqrt{\frac{2}{b}\hat{\sigma}^2} t_{\frac{\alpha}{2}}((a-1)(b-1)) \end{aligned}\right\} = 1 - \alpha$$

6.9

$$(1). 14.10 = \mu + \alpha_1 + \beta_1 + e_{11}, \quad 11.11 = \mu + \alpha_1 + \beta_2 + e_{12}, \dots,$$

$$14.10 = \mu + \alpha_3 + \beta_4 + e_{34}$$

(2)

ANOVA Table

		Sum of Squares	df	Mean Square	F	Sig.
Y * N	Between Groups	29.304	2	14.652	1.768	.225
	Within Groups	74.587	9	8.287		
	Total	103.891	11			

表明浓度对产品的得率的影响没有显著差异

(3)

ANOVA Table

		Sum of Squares	df	Mean Square	F	Sig.
Y * T	Between Groups	7.147	3	2.382	.197	.896
	Within Groups	96.744	8	12.093		
	Total	103.891	11			

表明温度对产品的得率的影响没有显著差异.

第四章习题提要

4. 1

$$X = \begin{pmatrix} 1 & 0 \\ 2 & -1 \\ 1 & 2 \end{pmatrix}, \quad X'X = \begin{pmatrix} 6 & 0 \\ 0 & 5 \end{pmatrix}, \quad (X'X)^{-1} = \begin{pmatrix} \frac{1}{6} & 0 \\ 0 & \frac{1}{5} \end{pmatrix}, \quad A = (1 \quad -2), \quad b = 0$$

$$A(X'X)^{-1}A' = 0.967$$

$$\hat{\beta} = \begin{pmatrix} \frac{y_1}{6} + \frac{y_2}{3} + \frac{y_3}{6} \\ -\frac{y_2}{5} + \frac{2y_3}{5} \end{pmatrix}$$

$$RSS = Y'Y - \hat{\beta}'X'Y = \frac{5y_1^2}{6} + \frac{2y_2^2}{15} + \frac{y_3^2}{30} - \frac{2y_1y_2}{3} - \frac{y_1y_3}{3} + \frac{2y_2y_3}{15}$$

$$(A\hat{\beta})^2 = \left(\frac{y_1}{6} + \frac{11y_2}{15} - \frac{19y_3}{30} \right)^2$$

$$F_H = \frac{30}{29} \frac{\left(\frac{y_1}{6} + \frac{11y_2}{15} - \frac{19y_3}{30} \right)^2}{\frac{5y_1^2}{6} + \frac{2y_2^2}{15} + \frac{y_3^2}{30} - \frac{2y_1y_2}{3} - \frac{y_1y_3}{3} + \frac{2y_2y_3}{15}}$$

4. 2

与例 4.1.2 类似

$$y_1 = X_1\beta_1 + e_1, e_1 \sim N(0, \sigma^2 I_n),$$

$$y_2 = X_2\beta_2 + e_2, e_2 \sim N(0, \sigma^2 I_m)$$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} X_1 & 0 \\ 0 & X_2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}, \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} \sim N(0, \sigma^2 I_{n+m}),$$

$$H: \begin{pmatrix} 0 & 1 & :0 & 1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = 0$$

$$\hat{\beta}_1 = \left(X_1' X_1 \right)^{-1} X_1' y_1, \quad \hat{\beta}_2 = \left(X_2' X_2 \right)^{-1} X_2' y_2$$

$$RSS = y_1' y_1 + y_2' y_2 - \hat{\beta}_1' X_1' y_1 - \hat{\beta}_2' X_2' y_2$$

约简模型为

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1_n & 0_n & \tilde{X}_1 \\ 0_m & 1_m & \tilde{X}_2 \end{pmatrix} \begin{pmatrix} \beta_{10} \\ \beta_{20} \\ \beta_{11} \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}, \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} \sim N(0, \sigma^2 I_{n+m}),$$

$$\text{其中: } \tilde{X}_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{1n} \end{pmatrix}, \quad \tilde{X}_2 = \begin{pmatrix} x_{21} \\ \vdots \\ x_{2m} \end{pmatrix}$$

$$\hat{\beta}_H = \begin{pmatrix} n & 0 & \sum_{i=1}^n x_{1i} \\ 0 & m & \sum_{i=1}^m x_{2i} \\ \sum_{i=1}^n x_{1i} & \sum_{i=1}^m x_{2i} & \sum_{i=1}^n x_{1i} + \sum_{i=1}^m x_{2i} \end{pmatrix}^{-1} \begin{pmatrix} \sum_{i=1}^n y_{1i} \\ \sum_{i=1}^m y_{2i} \\ \sum_{i=1}^n x_{1i} y_{1i} + \sum_{i=1}^m x_{2i} y_{2i} \end{pmatrix}$$

$$RSS_H = \sum_{i=1}^n y_{1i}^2 + \sum_{i=1}^m y_{2i}^2 - \hat{\beta}_H' \begin{pmatrix} \sum_{i=1}^n y_{1i} \\ \sum_{i=1}^m y_{2i} \\ \sum_{i=1}^n x_{1i} y_{1i} + \sum_{i=1}^m x_{2i} y_{2i} \end{pmatrix}$$

$$F_H = \frac{(RSS_H - RSS)/1}{RSS/(n+m-4)}$$

4.3

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}$$

$$\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}^{-1} \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}' \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$= \frac{1}{n \sum x_i^2 - (\sum x_i)^2} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix} \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix} = \begin{pmatrix} \frac{\sum x_i^2 \sum y_i - \sum x_i \sum x_i y_i}{n \sum x_i^2 - (\sum x_i)^2} \\ \frac{-\sum x_i \sum y_i + n \sum x_i y_i}{n \sum x_i^2 - (\sum x_i)^2} \end{pmatrix}$$

$$RSS = y' y - \hat{\beta}' X' y$$

$$= \sum y_i^2 - \frac{\sum x_i^2 \sum y_i - \sum x_i \sum x_i y_i}{n \sum x_i^2 - (\sum x_i)^2} \sum y_i + \frac{-\sum x_i \sum y_i + n \sum x_i y_i}{n \sum x_i^2 - (\sum x_i)^2} \sum x_i y_i$$

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \beta_H + \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}$$

$$\hat{\beta}_H = \frac{\sum x_i y_i}{\sum x_i^2}$$

$$RSS_H = y' y - \hat{\beta}_H' \tilde{X}' y = \sum y_i^2 - \frac{(\sum x_i y_i)^2}{\sum x_i^2}$$

$$F_H = \frac{(RSS_H - RSS)/1}{RSS/(n-2)}$$

4.4

$$RSS = y' y - \hat{\beta} X' y$$

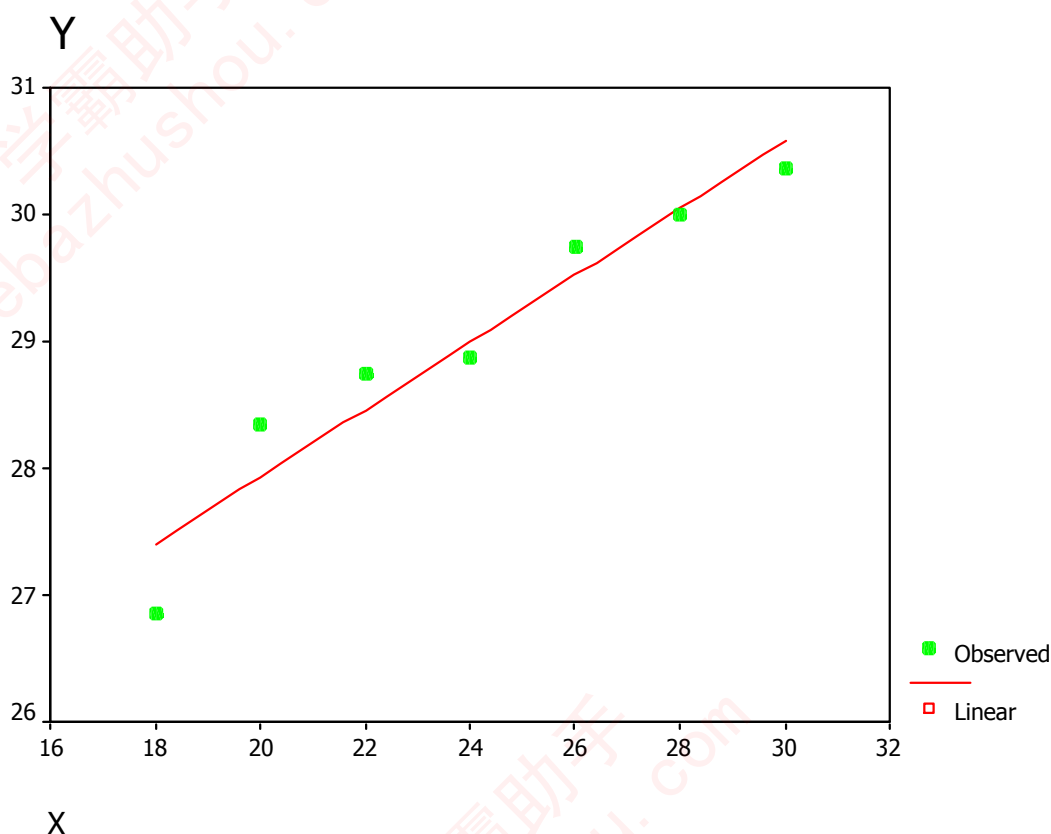
$$RSS_H = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$F_H = \frac{(RSS_H - RSS)/(p-1)}{RSS/(n-p)}$$

4.5

(1):

Listwise Deletion of Missing Data					
Multiple R		.95973			
R Square		.92109			
Adjusted R Square		.90530			
Standard Error		.36612			
Analysis of Variance:					
	DF	Sum of Squares	Mean Square		
Regression	1	7.8228571	7.8228571		
Residuals	5	.6702286	.1340457		
F =	58.35962	Signif F = .0006			
----- Variables in the Equation -----					
Variable	B	SE B	Beta	T	Sig T
X	.264286	.034595	.959732	7.639	.0006
(Constant)	22.648571	.841741		26.907	.0000



(2): $f(27.5)=29.91643$, 置信区间为: (28.86325, 30.96960)

4.6

(1)

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 360 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -1 & -1 \end{pmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{pmatrix}$$

$$\begin{pmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \\ \hat{\theta}_3 \end{pmatrix} = \left(\begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ -1 & -1 & -1 & \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right)^{-1} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \left(\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \\ 360 \end{pmatrix} \right)$$

$$\begin{pmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \\ \hat{\theta}_3 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}^{-1} \begin{pmatrix} y_1 - y_4 + 360 \\ y_2 - y_4 + 360 \\ y_3 - y_4 + 360 \end{pmatrix} = \begin{pmatrix} \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{pmatrix} \begin{pmatrix} y_1 - y_4 + 360 \\ y_2 - y_4 + 360 \\ y_3 - y_4 + 360 \end{pmatrix}$$

$$= \frac{1}{4} \begin{pmatrix} 3y_1 - y_2 - y_3 - y_4 + 360 \\ 3y_2 - y_1 - y_3 - y_4 + 360 \\ 3y_3 - y_1 - y_2 - y_4 + 360 \end{pmatrix}$$

$$\hat{\theta}_4 = 360 - \hat{\theta}_1 - \hat{\theta}_2 - \hat{\theta}_3 = \frac{1}{4}(3y_4 - y_1 - y_2 - y_3 + 360)$$

$$RSS = \sum_{i=1}^3 y_i^2 + (y_4 - 360)^2 - \begin{pmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \\ \hat{\theta}_3 \end{pmatrix}' \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \\ 360 \end{pmatrix}$$

$$= \sum_{i=1}^3 y_i^2 + (y_4 - 360)^2 - \frac{1}{4}(3y_1 - y_2 - y_3 - y_4 + 360)(y_1 - y_4 + 360)$$

$$- \frac{1}{4}(3y_2 - y_1 - y_3 - y_4 + 360)(y_2 - y_4 + 360) - \frac{1}{4}(3y_3 - y_1 - y_2 - y_4 + 360)(y_3 - y_4 + 360)$$

(2)

$$H: \theta_1 = \theta_3, \theta_2 = 360 - \theta_1 - \theta_2 - \theta_3 \Leftrightarrow H: \theta_1 = \theta_3, \theta_2 = 180 - \theta_3$$

约简模型为

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 180 \\ 0 \\ 180 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} \theta_3 + \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{pmatrix}$$

$$\hat{\theta}_{3H} = (\tilde{X}\tilde{X})^{-1} \tilde{X}\tilde{Y} = \frac{1}{4}(y_1 - (y_2 - 180) + y_3 - (y_4 - 180)) = \frac{1}{4}(360 + y_1 - y_2 + y_3 - y_4)$$

$$RSS_H = y_1^2 + y_3^2 + (y_2 - 180)^2 + (y_4 - 180)^2 - \hat{\theta}_{3H} \tilde{X}\tilde{Y}$$

$$= y_1^2 + y_3^2 + (y_2 - 180)^2 + (y_4 - 180)^2$$

$$- \frac{1}{4}(360 + y_1 - y_2 + y_3 - y_4)(y_1 - (y_2 - 180) + y_3 - (y_4 - 180))$$

$$= y_1^2 + y_3^2 + (y_2 - 180)^2 + (y_4 - 180)^2 - \frac{1}{4}(360 + y_1 - y_2 + y_3 - y_4)^2$$

$$F_H = \frac{(RSS_H - RSS)/2}{RSS/(4-3)} \sim F(2,1)$$

4. 7

$$\tilde{Y} = \frac{1}{d_0}(Y - c_0 \mathbf{1}_n) \quad , \quad \tilde{X} = (X - \mathbf{1}_n \mathbf{1}_{p-1}' C) D^{-1} \quad , \quad \text{其中} \quad C = \begin{pmatrix} c_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & c_{p-1} \end{pmatrix} \quad ,$$

$$\tilde{X}\tilde{Y} = D^{-1}\left(X' - C'\mathbf{1}_{p-1}\mathbf{1}_n'\right)\frac{1}{d_0}(Y - c_0\mathbf{1}_n) = \frac{1}{d_0}D^{-1}\left(XY - C'\mathbf{1}_{p-1}\mathbf{1}_n'Y - c_0X'\mathbf{1}_n + c_0nC'\mathbf{1}_{p-1}\right)$$

$$\begin{aligned}\tilde{\beta} &= (\tilde{X}\tilde{X})^{-1}\tilde{X}\tilde{Y} \\ &= \frac{1}{d_0}D(X'X - X'\mathbf{1}_n\mathbf{1}_{p-1}'C - C'\mathbf{1}_{p-1}\mathbf{1}_n'X + nC'\mathbf{1}_{p-1}\mathbf{1}_{p-1}'C)^{-1}(X'Y - C'\mathbf{1}_{p-1}\mathbf{1}_n'Y - c_0X'\mathbf{1}_n + c_0nC'\mathbf{1}_{p-1})\end{aligned}$$

$$\begin{aligned} \tilde{\beta} &= (\tilde{X}'\tilde{X})^{-1} \tilde{X}'\tilde{Y} \\ &= \frac{1}{d_0} D(X'X - X'\mathbf{1}_n \mathbf{1}_{p-1}' C - C'\mathbf{1}_{p-1} \mathbf{1}_n' X + nC'\mathbf{1}_{p-1} \mathbf{1}_{p-1}' C)^{-1} X'X\hat{\beta} \\ &\quad + \frac{1}{d_0} D(X'X - X'\mathbf{1}_n \mathbf{1}_{p-1}' C - C'\mathbf{1}_{p-1} \mathbf{1}_n' X + nC'\mathbf{1}_{p-1} \mathbf{1}_{p-1}' C)^{-1} (-C'\mathbf{1}_{p-1} \mathbf{1}_n' Y - c_0 X'\mathbf{1}_n + c_0 nC'\mathbf{1}_{p-1}) \end{aligned}$$

$$RSS\tilde{S} = \tilde{Y}'\tilde{Y} - \tilde{\beta}'\tilde{X}'\tilde{Y} = \frac{1}{d_0^2}(Y - c_0\mathbf{1}_n)'(Y - c_0\mathbf{1}_n) \\ - \frac{1}{d_0^2}(XY - C'\mathbf{1}_{p-1}\mathbf{1}_n'Y - c_0X'\mathbf{1}_n + c_0nC'\mathbf{1}_{p-1})'(XX - X'\mathbf{1}_n\mathbf{1}_{p-1}'C - C'\mathbf{1}_{p-1}\mathbf{1}_n'X + nC'\mathbf{1}_{p-1}\mathbf{1}_{p-1}'C)^{-1} \\ (XY - C'\mathbf{1}_{p-1}\mathbf{1}_n'Y - c_0X'\mathbf{1}_n + c_0nC'\mathbf{1}_{p-1})$$

????????????????????????????????很繁！没完！

4. 8

$$(1) \quad X = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \quad \mathbf{x}_0 = \begin{pmatrix} 1 \\ x_0 \end{pmatrix}$$

$$X'X = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$$

$$(X'X)^{-1} = \frac{1}{n \sum x_i^2 - (\sum x_i)^2} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix}$$

$$\begin{aligned} \mathbf{x}_0' (X'X)^{-1} \mathbf{x}_0 &= \frac{\sum x_i^2 - 2x_0 \sum x_i + nx_0^2}{n \sum x_i^2 - (\sum x_i)^2} = \frac{1}{n} + \frac{\frac{1}{n} (\sum x_i)^2 - 2x_0 \sum x_i + nx_0^2}{n \sum x_i^2 - (\sum x_i)^2} \\ &= \frac{1}{n} + \frac{n\bar{x}^2 - 2nx_0\bar{x} + nx_0^2}{n \sum x_i^2 - n^2(\bar{x})^2} = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \end{aligned}$$

$$Var(\hat{y}_0 - y_0) = \sigma^2 \left(1 + \mathbf{x}_0' (X'X)^{-1} \mathbf{x}_0 \right) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right)$$

$$(2) \quad Var(\hat{\mu}_0) = Var(\mathbf{x}_0' \hat{\beta}) = \sigma^2 \left(\mathbf{x}_0' (X'X)^{-1} \mathbf{x}_0 \right) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right)$$

4. 9

$$Y = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \beta + \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}$$

$$(1) \quad \hat{\beta} = (X'X)^{-1} X'Y = \frac{\sum x_i y_i}{\sum x_i^2}$$

$$\hat{\sigma}^2 = Y'Y - \hat{\beta}'X'Y = \sum y_i^2 - \frac{(\sum x_i y_i)^2}{\sum x_i^2}$$

$$(2) \quad Var(\hat{y}_0 - y_0) = \sigma^2 \left(1 + \mathbf{x}_0' (X'X)^{-1} \mathbf{x}_0 \right) = \sigma^2 \left(1 + \frac{x_0^2}{\sum x_i^2} \right)$$

$$(3) \quad Var(\hat{\mu}_0) = Var(\mathbf{x}_0' \hat{\beta}) = \sigma^2 \left(\mathbf{x}_0' (X'X)^{-1} \mathbf{x}_0 \right) = \sigma^2 \left(\frac{x_0^2}{\sum x_i^2} \right)$$